Order Monotremata

Oviparity (egg-laying) is the distinctive reproductive feature of the highly specialized monotremes. The yolk-filled ova, already large at fertilization (4 mm in diameter), acquire a permeable shell membrane and shell through which absorption of nutritive, uterine fluids occurs (5078). Cleavage is meroblastic initially (1550). The 15x17 mm, ovoid eggs are laid when the embryos have 19-20 somites (3358, 3360, 3361, 4178, 4738, 4768, 5079, 11761, 11901). Embryogenesis occurs mainly during incubation. Although gestation and lactation lengths appear to be longer in the larger echidna than in the platypus, incubation lengths are similar, as is the time to sexual maturity.

Hatchlings are tiny, naked, and blind with a specialized, deciduous, egg-tooth bone and well-developed olfaction (1055, 4069). They suckle from mammary areolae which are similar to those of humans (4064). Echidna females possess a pouch, while platypus females do not. Lactation is long and milk composition is variable, but usually high in total solids, lipids, iron, and copper (4065, 4067, 4069, 5288, 7247a, 8245a, 10724).

The name of the order refers to the presence of a cloaca which communicates via a single vagina with a pair of uteri in females. Males have abdominal testes and epididymides; two types of accessory glands; a bifid or lobate penis within the cloaca; a subdivided urethra with multiple exits; and filiform spermatozoa (709, 1621, 2641, 4064). Urine does not pass through the penis (5431a). A spur is often present on the hind ankle of adult males and may be present on juveniles of both sexes (4064, 10474, 11152). The presence of a Y-chromosome is controversial (4013). Monotremes do not have antral follicles, bacula, scrota, or seminal vesicles.

Reviews of the general biology of monotremes (449a, 449b, 4064, 4064a) and captive breeding (1987, 3721) are available.

Order Monotremata, Family Tachyglossidae

Echidnas (*Tachyglossus* 2.5-6 kg; *Zaglossus* 5-10 kg) are specialists on ants and termites or earthworms in Australia and New Guinea. Both uteri and ovaries are functional (3358, 3360, 4768, 5078, 8164). Penial spines are not present (1031; cf 1621). Females have one more chromosome than males (4013). Details of female anatomy (4065, 4069, 8162, 8164), male anatomy (709, 2638, 2639, 2640, 4945, 5431a, 11101), endocrinology (9789a), sperm maturation (2641, 5431a), mating (4512b), lactation (4028, 4063, 4068), and milk (5445a, 5505, 6865, 7243a, 10725) are available.

HATCHLING MASS[1]		
Tachyglossus aculeatus	378 mg	4069
	380 mg	321
HATCHLING SIZE		
Tachyglossus aculeatus	12 mm	10698
	12.5 mm	6664
	14.7 mm	4069
Zaglossus bruijni	17 mm	11151
WEANING MASS		
Tachyglossus aculeatus	400 g; n1; pouch exit	6063
Zaglossus bruijni	~400 g; pouch exit	11151
WEANING SIZE		
Zaglossus bruijni	305 mm; pouch exit	11151
LITTER SIZE		
Tachyglossus aculeatus	1 egg	11901
	1 egg	9763
	1.03; r1-2; n140 lt	3358, 3360
	2 eggs; n1 f	1338
	2 eggs; r1-3; n2 f	8164
Zaglossus bruijni	mode 1 egg; r1-3	11151

[1] The left column lists, in CAPITAL letters, the reproductive variables and, in *italics*, the scientific names of the species. The middle column provides the data, and the right column gives the citations to the publications in which the data were found.

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WEANING SIZE		
Zaglossus bruijni	305 mm; pouch exit	11151
LITTER SIZE		
Tachyglossus aculeatus	1 egg	11901
	1 egg	9763
	1.03; r1-2; n140 lt	3358, 3360
	2 eggs; n1 f	1338
	2 eggs; r1-3; n2 f	8164
Zaglossus bruijni	mode 1 egg; r1-3	11151

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SEXUAL MATURITY		
Tachyglossus aculeatus	~1 yr	8798
	est 2 yr	777
CYCLE LENGTH		
Tachyglossus aculeatus	est 9-27 d	10698
	>16 d	4069
	est 18-28 d; n1	1338
	1 mo	8798
GESTATION		
Tachyglossus aculeatus	min 17 d; removal of male to birth	4064, 4064a
	est 18-27 d	9142a
	27 d	1338
INCUBATION		
Tachyglossus aculeatus	10 d	9142a
	10-10.5 d	4069
Zaglossus bruijni	~10 d	11151
LACTATION		
Tachyglossus aculeatus	pouch exit: ~55 d	4064
	pouch exit: 2 mo	10698
	pouch exit: 63 d	4027
	pouch exit: 3 mo	10474
	wean: 3+ mo after pouch exit	4064
	independent: 7-9 mo	10698
	wean: 195-200 d; n1	17
	wean: ~200 d	4068
Zaglossus bruijni	pouch exit: 8 wk	11151
	matemal care: 3 mo	11151
	wean: 5 mo	10809
INTERLITTER INTERVAL		
Tachyglossus aculeatus	est 2 yr	777
SEASONALITY		
Tachyglossus aculeatus	estrus: end July; Australia	9763
	mate: June-early Sept (last egg); TAS, Australia	3360
	mate: July-Aug; Australia	10474
	mate: July-Aug; Snowy Mt, NSW, Australia	680a
	mate: est early Sept; n1; Australia	1338
	mate: mid Sept; SA, Australia	11901
	mate: start rainy season; n Australia	2214
	eggs in utero: end July; se Australia	1550
	eggs in utero: Sept-Oct; QLD, Australia	777
	eggs laid: early Oct; n1; Australia	1338
	pouch with egg: end Aug; Australia	9763

Order Monotremata, Family Ornithorhynchidae, Genus *Ornithorhynchus*

The platypus (*Ornithorhynchus anatinus* 0.5-2 kg) is a highly specialized, aquatic, nocturnal predator of invertebrates in streams of temperate Australia. Males may be larger than females (8798). Penial spines are present (1031). The left ovary and uterus are prominent and preferentially used (775, 3360, 3607, 4768, 5078, 8157, 8158, 8159, 9763). Eyes open at 11+ wk (3302). Details of anatomy (3615, 3627, 4767, 4778, 4944, 8157, 11101), endocrinology (1618, 1619, 1620, 1621, 1767, 4315a, 7049, 7059a), courtship (10474a), milk composition (222, 3764, 7247), and reproductive energetics (4028a) are available.

HATCHLING SIZE		
Ornithorhynchus anatinus	25 mm	10698
	~25 mm	8798
LITTER SIZE		
Ornithorhynchus anatinus	1.83 eggs in utero; r1-2; n6 f	4737
	2 oocytes	4768
	2 eggs	11901
	2 eggs; r1-3	8798
	mode 2; r1-4	775
	2 eggs in utero; n1 f	1550

	2 eggs in utero; n1 f	4772
	2 eggs in nest?	1467
	2 yg ea; n2 nest	776
	2.0; n2 lt	11761
	2 cl ea; n2 f	8158
	2.5; r2-3; n2	775
	3 cl; n1 ovary	8159
	3 emb; n1 lt	8159
SEXUAL MATURITY		
Ornithorhynchus anatinus	f: smallest lact f: 670-750 g; n4 areas	4010
	f: min 2nd season	4009a
	f/m: 1st mate: 1-2 yr	4010
	f/m: 1 yr	8798
	f/m: 2 yr	9763
GESTATION		
Ornithorhynchus anatinus	>10 d	5078
	12-14 d	10698
	est 14 d	3302, 3303
	1 mo	8798
INCUBATION		
Ornithorhynchus anatinus	incubate: est 7-10 d	3302
	incubate: 1-2 wk	10474
	incubate: 12 d	3303
LACTATION		
Ornithorhynchus anatinus	wean: 3-4 mo; field data	4009, 4009a, 4010
	den emergence: 4 mo	8798
	nurse: 4-5 mo	10474
INTERLITTER INTERVAL		
Ornithorhynchus anatinus	1 yr	4010
SEASONALITY		
Ornithorhynchus anatinus	quiescent: May-July; NSW, Australia	4010
	estrus: mid Aug; Australia	9763
	mate: Aug; QLD, Australia	10474
	mate: Aug-Sept; Australia	1467
	mate: Aug-Sept; NSW, Australia	3607
	mate: Sept; NSW, VIC, Australia	10474
	mate: Oct; TAS, Australia	10474
	mate: Oct; 1 pr; captive	3303
	f with ova: Oct; n3; Australia	8158
	uterine eggs: end Aug; Australia	9763
	uterine eggs: Aug-Sept; n6; Australia	4737
	uterine eggs: mid Aug; n1; NSW, Australia	1550
	uterine eggs: Oct; NSW, Australia	775
	uterine eggs: Oct; n1; Australia	4772
	lay eggs: mid Aug-Oct; Australia	1467
	lay eggs: end July-early Sept; TAS, e Australia	8798
	hatch: Oct; VIC, Australia	5701
	lact f: Oct-Mar, peak Dec; NSW, Australia	4010
	lact f: Dec; Australia	8162
	young found: Oct-Nov; n2 nests; Australia	776